DATABASE MANAGEMENT SYSTEMS (CS301)

Complete Syllabus

Course Description

Database Management Systems (DBMS) deals with the design, implementation, and management of databases. The course covers data models, database architecture, SQL, normalization, transactions, concurrency control, and recovery mechanisms.

Course Objectives

• Understand database concepts and architecture.
• Learn data modeling using ER diagrams.
• Develop SQL query writing skills.
• Apply normalization techniques.
• Understand transaction management and concurrency control.
• Study recovery mechanisms for database reliability.

Unit 1 - Introduction to Databases

Introduction to Databases, Database Management Systems, Advantages of DBMS, Database Users, Database Architecture, Data Models, and Three-Level Architecture.

Unit 2 - ER Model & Relational Model

Entity Relationship Model, Entities, Attributes, Relationships, Cardinality Constraints, ER Diagrams, Relational Model Concepts, and Keys.

Unit 3 - SQL Queries

SQL Basics, DDL Commands, DML Commands, DCL Commands, TCL Commands, Joins, Nested Queries, Aggregate Functions, and Stored Procedures.

Unit 4 - Functional Dependencies & Normalization

Functional Dependencies, Types of Dependencies, Normalization, First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), and Boyce-Codd Normal Form (BCNF).

Unit 5 - Transactions & Recovery

Transactions, ACID Properties, Concurrency Control, Locking Mechanisms, Deadlocks, Recovery Techniques, Logging, and Checkpointing.

Course Outcomes

CO1: Explain fundamental database concepts and architecture.
CO2: Design ER diagrams and relational schemas.
CO3: Develop SQL queries for database operations.
CO4: Apply normalization techniques to improve database design.
CO5: Analyze transaction management and recovery mechanisms.

Assessment Pattern

Internal assessments, assignments, laboratory exercises, quizzes, and semester-end examinations.

Recommended Textbooks

1. Database System Concepts – Silberschatz, Korth.
2. Fundamentals of Database Systems – Elmasri & Navathe.

Summary

This course provides a strong foundation in database design, implementation, querying, and transaction management, which are essential skills for software development and data management.